

“Job or Jab?”: Detecting Fake Job Postings Using NLP

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Abstract— This study focuses on detecting fraudulent job postings using Natural Language Processing (NLP) and a comparative analysis of six machine learning models: Logistic Regression, Random Forest, k-Nearest Neighbors, Multilayer Perceptron, Support Vector Machine, and Naive Bayes. The dataset underwent text preprocessing and feature transformation using TF-IDF, with ADASYN applied to address class imbalance. A 70:20:10 data split was used for training, testing, and validation, respectively. Evaluation metrics including accuracy, precision, recall, and confusion matrices were generated for each model. Among the models tested, the Multilayer Perceptron yielded the highest accuracy, making it the most effective in classifying job postings as real or fake. This research demonstrates that machine learning, combined with proper sampling techniques and preprocessing, can significantly improve the reliability of fake job detection systems and help protect users from online recruitment fraud.

Keywords—Machine Learning, Natural Language Processing (NLP), Fake Job Detection, Logistic Regression, Support Vector Machine, TF-IDF, Dataset Imbalance

I. INTRODUCTION

In this generation, almost everything is being handled by technology, such as online banking, education, security, job, etc. Scammers make use of technology to increase their advantage to steal from people to get money fast [1]. Job seekers. Especially those new to the workforce, are particularly vulnerable to online job scams that appear professional and credible, making it crucial to deploy technologies that can proactively identify such deceptive practices. According to the Department of Migrant Workers (DMW), there are over 71,000 fake job postings in social media platforms they have taken down in 2024 and in collaboration with Meta Philippines and TikTok Philippines, they have made possible the deletion of 71,653 fake job postings and accounts in the same year, including the 50,220 suspicious posts on Facebook and 21,433 on TikTok [2]. To address this threat, this study proposes the implementation of Natural Language Processing (NLP) and Machine Learning (ML) to detect and predict fake job postings, which aims to

help job seekers make more informed, safe, and confident decisions when applying for jobs online.

Fake information, including job scams, has evolved alongside technology, with researchers noting that fraudulent online content has become increasingly sophisticated and harder to detect [3]. Scammers often mimic legitimate job advertisements by replicating the language and format used in real postings, making it difficult for both users and basic detection systems to differentiate them [4]. Studies have demonstrated the effectiveness of Natural Language Processing (NLP) techniques, particularly TF-IDF and Bag-of-Words, in extracting features from deceptive text and improving classification accuracy [5]. Research has also emphasized the importance of detecting opinion spam using supervised learning—a concept directly applicable to identifying fake job posts based on linguistic patterns [6]. These previous works provide the foundation for our present study, which builds upon their methodologies using modern NLP and machine learning techniques to detect fraudulent job listings on online platforms.

Despite the increasing prevalence of online job scams, there remains a limited number of research studies focused on the automatic detection of fraudulent job postings using advanced machine learning techniques. Existing approaches often struggle with the challenge of highly imbalanced datasets, where the number of legitimate job postings significantly outweighs the number of fraudulent ones. This imbalance causes poor classification performance, particularly in identifying the minority class of fake jobs. Moreover, many models fail to integrate robust Natural Language Processing (NLP) methods that can effectively analyze the linguistic patterns used in scam postings. Therefore, there is a pressing need for more comprehensive frameworks that combine NLP and machine learning, along with advanced sampling techniques, to improve the accuracy and reliability of fake job post detection systems [7].

This study aims to develop a system that detects fraudulent job postings using machine learning and natural language processing techniques. It incorporates methods such as data mining, programming, data engineering, and

visualization to analyze and classify job listings. Specifically, these are the objectives:

- To clean, preprocess, and manipulate the dataset effectively to ensure high-quality model input.
- To apply NLP techniques such as TF-IDF to extract relevant textual features from job descriptions.
- To address dataset imbalance using sampling techniques such as ADASYN to improve model accuracy.
- To create a machine learning-based model that accurately classifies job postings as legitimate or fake.
- To implement and compare multiple supervised machine learning algorithms (e.g., Logistic Regression, SVM, Random Forest, k-NN, Multilayer Perceptron, Naïve Bayes).
- To visualize and interpret the results using appropriate data visualization tools and performance metrics.

This study emphasizes the importance of safeguarding job seekers in an increasingly digital job market. Scammers use the wide reach and accessibility of online job platforms to deceive applicants, compromising their financial security and personal information. Fake job postings also affect the credibility and trustworthiness of recruitment websites, discouraging users from seeking legitimate opportunities. Implementing NLP and ML-based systems for automatic detection of fraudulent listings can empower job seekers to make informed, safe decisions. Furthermore, this research aims to contribute to building a more secure online job-seeking environment and to restore trust in digital employment platforms.

Moreover, this study focuses on detecting fraudulent job postings using Natural Language Processing (NLP) and supervised Machine Learning (ML) techniques, utilizing a publicly available dataset from Kaggle that contains approximately 18,000 job postings, including around 800 labeled as fake [8]. The project involves data preprocessing, feature extraction using TF-IDF, and the application of classification models such as Support Vector Machine, Random Forest, Logistic Regression, and others, with the ADASYN technique used to handle class imbalance. However, the study is limited by the dataset's lack of diversity, potential overfitting due to the small number of fake postings, and its reliance on historical data rather than real-time job listings. The model's accuracy is also constrained by the quality of the textual input, and it may struggle to detect more recent or sophisticated scam formats. Additionally, the study does not include live web integration or deep learning-based approaches, which could further enhance detection accuracy in real-world scenarios.

II. METHODOLOGY

This study implements a machine learning-based approach to detect fraudulent job postings by leveraging natural language processing (NLP) techniques. The methodology is structured into six major stages: data preprocessing, feature engineering, exploratory analysis, model development, evaluation, and interpretability.

A. Conceptual Framework

The conceptual framework illustrates the systematic six-stage methodology employed in developing a machine learning-based system for detecting fraudulent job postings. The process flows sequentially from Data Preprocessing (cleaning the Kaggle dataset and handling missing values), through Feature Engineering (applying TF-IDF vectorization to transform textual data into numerical features), to Exploratory Data Analysis (EDA) for visual pattern identification using word clouds. The methodology then proceeds to Model Development, where six supervised learning algorithms are trained using ADASYN oversampling to address class imbalance, followed by comprehensive Evaluation & Testing using standard performance metrics across 70:20:10 data splits. The framework concludes with Model Interpretability analysis to identify the most significant linguistic indicators contributing to fraud detection, ensuring the explainability and trustworthiness of the classification system.

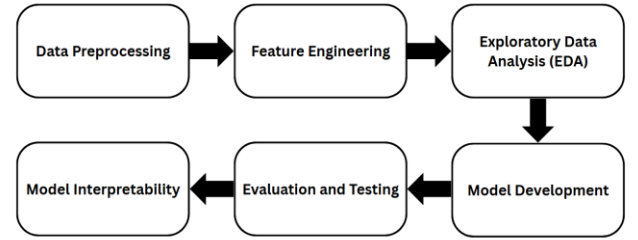


Fig. 1. Conceptual Framework for Fake Job Posting Detection.

B. Data Preprocessing

The dataset used in this study is the Fake Job Postings Prediction dataset from Kaggle, consisting of approximately 18,000 job advertisements labeled as either real (0) or fake (1). Initial preprocessing involved removing records with missing values in critical fields such as description. Missing values in non-critical binary fields such as `has_company_logo` and `telecommuting` were filled with default values (1 and 0 respectively) to preserve dataset size.

Additionally, multiple text fields (title, `company_profile`, description, requirements, and benefits) were concatenated into a single composite feature named `text`. This enrichment was designed to provide the model with a more complete semantic context per job listing.

C. Feature Engineering

To prepare the textual data for machine learning, the `text` field underwent tokenization and was vectorized using Term Frequency-Inverse Document Frequency (TF-IDF), a well-

established technique for weighting words based on their importance in a document relative to the corpus. The vectorizer was limited to the top 5,000 features to reduce dimensionality and improve performance.

D. Exploratory Data Analysis

Descriptive analysis was conducted to visualize patterns and differences between real and fake job postings. Word clouds were generated separately for both classes to reveal frequently occurring terms. This step served both to validate class separation heuristically and to provide qualitative insight into common characteristics of fake listings (e.g., vague language, overpromising benefits).

E. Model Development

Six supervised machine learning models were evaluated for this study: Logistic Regression, Random Forest, k-Nearest Neighbors (k-NN), Multilayer Perceptron (MLP), Support Vector Machine (SVM), and Naïve Bayes. Each model was trained using the TF-IDF vectors as input features and the fraudulent label as the target variable.

To mitigate the effect of class imbalance, class weights were computed and applied using sklearn’s `compute_class_weight()` function with ADASYN oversampling.

The dataset was split into training, testing, and validation subsets using a 70:20:10 ratio, with a fixed random seed to ensure reproducibility. After comprehensive evaluation, the top three models—Multilayer Perceptron, Random Forest, and Support Vector Machine—demonstrated the highest performance and are presented in detail in the Results section.

F. Evaluation and Testing

Model performance was assessed on the test set using standard classification metrics: accuracy, precision, recall, and F1-score. These metrics were computed using `classification_report()`. A confusion matrix was also visualized to better understand the model's misclassification behavior, particularly its ability to correctly detect fraudulent postings.

G. Model Interpretability

To interpret the model’s decision-making process, the top 20 positive and negative TF-IDF features were extracted based on their logistic regression coefficients. Features with the highest positive coefficients are indicative of fake job postings, while those with the most negative coefficients are associated with legitimate job listings. These visualizations support the explainability of the model and enhance user trust.

III. RESULTS AND DISCUSSION

This section presents the findings from the analysis, model development, and evaluation stages, followed by a discussion of their implications, the effectiveness of the methodology, and the study's limitations.

A. Word Cloud Analysis

Visual exploration using word clouds provided initial insights into the linguistic patterns differentiating real and fake job postings.

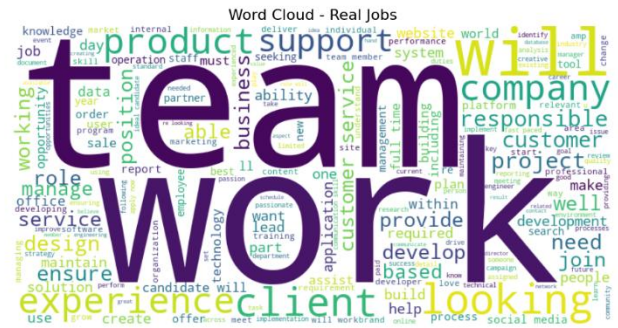


Fig. 2. Real Jobs Word Cloud.

The word cloud generated from legitimate job descriptions frequently highlighted terms such as "team work", "will", "experience", "looking", and "product support". These terms are characteristic of standard professional language, focusing on qualifications, responsibilities, and team-oriented work environments.



Fig. 3. Fake Job Word Cloud.

In contrast, the word cloud for fake job (Fig. 3) postings prominently featured words like "work", "product", "project", and "company". This vocabulary often suggests promises of easy money, remote work without clear prerequisites, and vague benefits, consistent with common scam tactics. These visual differences provided preliminary confirmation of distinct linguistic patterns between the two classes.

B. Model Performance Evaluation

After feature engineering using TF-IDF and splitting the data into 70% training, 20% testing, and 10% validation sets, the training data underwent ADASYN oversampling to address the class imbalance. This resulted in a balanced training set with 11,910 real jobs and 11,837 fake jobs. Six supervised machine learning models were trained and evaluated.

a. Model’s Performance Summary

Table 1 summarizes the Random Forest model’s performance on the Test Set. It shows very high overall accuracy (0.98) and near-perfect detection of real jobs. Crucially, for fake jobs, it achieves exceptionally high

precision (0.97) but a moderate recall (0.64), leading to a strong F1-score of 0.77. This suggests very few false alarms when identifying fake postings.

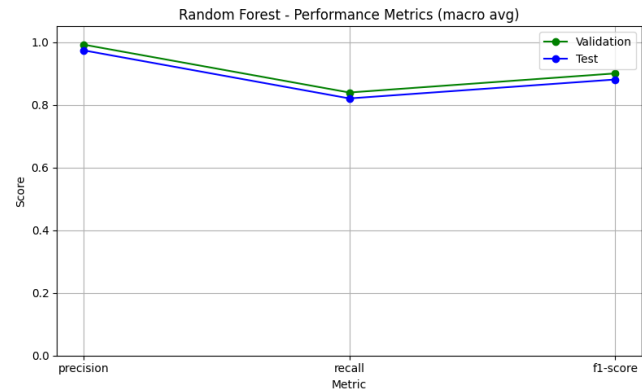


TABLE I. EVALUATION ON TEST SET FOR RANDOM FOREST

	Precision	Recall	F1-score	support
Real	0.98	1.00	0.99	3403
Fake	0.97	0.64	0.77	173
Accuracy			0.98	3576
Macro avg.	0.97	0.82	0.88	3576
Weight avg.	0.98	0.98	0.98	3576

Table 2 provides the Random Forest model's performance on the Validation Set. It maintains the strong performance seen on the Test Set, with perfect precision (1.00) and improved recall (0.68) for fake jobs, yielding an impressive F1-score of 0.81. This indicates high confidence in its fake job predictions.

TABLE II. EVALUATION ON VALIDATION SET FOR RANDOM FOREST

	Precision	Recall	F1-score	support
Real	0.98	1.00	0.99	1701
Fake	1.00	0.68	0.81	87
Accuracy			0.98	1788
Macro avg.	0.99	0.84	0.90	1788
Weight avg.	0.98	0.98	0.98	1788

Table 3 summarizes the Multilayer Perceptron (MLP) model's performance on the Test Set. It shows excellent overall accuracy (0.98) and strong, balanced performance for both classes. For fake jobs, it achieve high precision (0.83) and recall (0.77), resulting in the highest F1-score (0.80) for the minority class on this set, indicating a robust classifier.

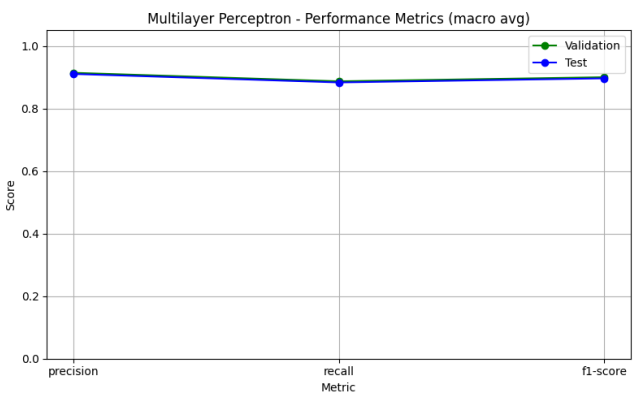


TABLE III. EVALUATION ON TEST SET FOR MULTILAYER PERCEPTRON

	Precision	Recall	F1-score	support
Real	0.99	0.99	0.99	3403
Fake	0.83	0.77	0.80	173
Accuracy			0.98	3576
Macro avg.	0.91	0.88	0.90	3576
Weight avg.	0.98	0.98	0.98	3576

Table 4 displays the Multilayer Perceptron model's performance on the Validation Set. The results remain consistently strong, with high precision (0.84) and recall (0.78) for fake jobs, leading to an F1-score of 0.81. This confirms MLP as a top-performing and well-balanced model for fake job detection.

TABLE IV. EVALUATION ON VALIDATION SET FOR MULTILAYER PERCEPTRON

	Precision	Recall	F1-score	support
Real	0.99	0.99	0.99	1701
Fake	0.84	0.78	0.81	87
Accuracy			0.98	1788
Macro avg.	0.91	0.89	0.90	1788
Weight avg.	0.98	0.98	0.98	1788

Table 5 shows the Support Vector Machine (SVM) model's performance on the Test Set. It achieves high overall accuracy (0.98) and excellent performance on both real and fake jobs. For fake jobs, it maintains strong precision (0.80) and good recall (0.77), resulting in a high F1-score of 0.79.

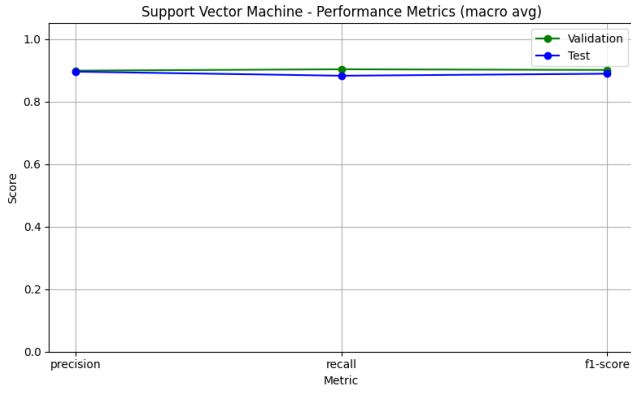


TABLE V. EVALUATION ON TEST SET FOR SUPPORT VECTOR MACHINE

	Precision	Recall	F1-score	support
Real	0.99	0.99	0.99	3403
Fake	0.80	0.77	0.79	173
Accuracy			0.98	3576
Macro avg.	0.90	0.88	0.89	3576
Weight avg.	0.98	0.98	0.98	3576

Table 6 presents the Support Vector Machine model's performance on the Validation Set. Its results are consistent with the Test Set, maintaining high precision (0.81) and recall (0.82) for fake jobs, leading to an F1-score of 0.81. This indicates SVM's consistent and robust performance.

TABLE VI. EVALUATION ON VALIDATION SET FOR SUPPORT VECTOR MACHINE

	Precision	Recall	F1-score	support
Real	0.99	0.99	0.99	1701
Fake	0.81	0.82	0.81	87
Accuracy			0.98	1788
Macro avg.	0.90	0.90	0.90	1788
Weight avg.	0.98	0.98	0.98	1788

Table 7 presents the Logistic Regression model's performance on the Test Set. It shows high overall accuracy (0.97) and excellent detection of real jobs. For fake jobs, it achieves a good recall (0.87) but with moderate precision (0.64), resulting in an F1-score of 0.74, indicating a decent balance for the minority class.

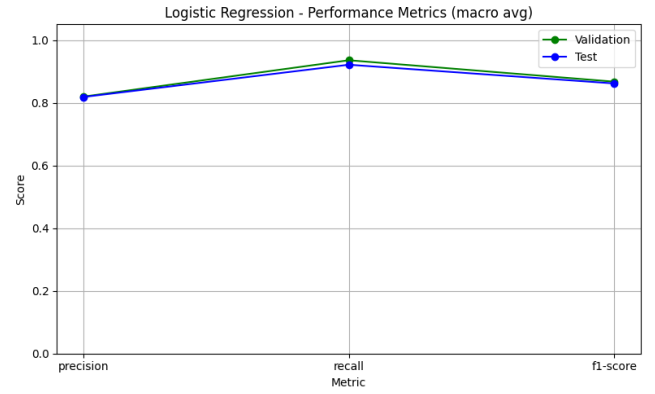


TABLE VII. EVALUATION ON TEST SET FOR LOGISTIC REGRESSION

	Precision	Recall	F1-score	support
Real	0.99	0.98	0.98	3403
Fake	0.64	0.87	0.74	173
Accuracy			0.97	3576
Macro avg.	0.82	0.92	0.86	3576
Weight avg.	0.98	0.97	0.97	3576

Table 8 displays the Logistic Regression model's performance on the Validation Set. The results are highly consistent with the Test Set, confirming the model's stable generalization. Performance on fake jobs is very similar (0.90 recall, 0.64 precision, 0.75 F1-score), indicating consistent ability to detect scams

TABLE VIII. EVALUATION ON VALIDATION SET FOR LOGISTIC REGRESSION

	Precision	Recall	F1-score	support
Real	0.99	0.97	0.98	1701
Fake	0.64	0.90	0.75	87
Accuracy			0.97	1788
Macro avg.	0.82	0.94	0.87	1788
Weight avg.	0.98	0.97	0.97	1788

Table 9 shows the k-Nearest Neighbors model's performance on the Test Set. While achieving very high recall (0.97) for fake jobs, meaning it catches most scams, its precision (0.15) for fake jobs is very low. This results in a poor F1-score (0.27), indicating a high rate of false positives and limited practical utility despite high overall accuracy for real jobs.

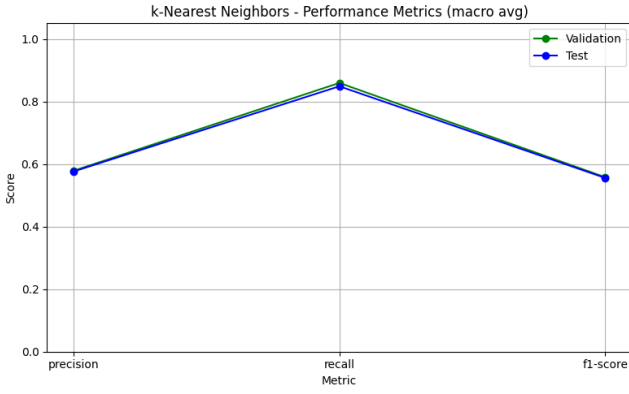


TABLE IX. EVALUATION ON TEST SET FOR K-NEAREST NEIGHBORS

	Precision	Recall	F1-score	support
Real	1.00	0.73	0.84	3403
Fake	0.15	0.97	0.27	173
Accuracy			0.74	3576
Macro avg.	0.58	0.85	0.56	3576
Weight avg.	0.96	0.74	0.82	3576

Table 10 presents the k-Nearest Neighbors model's performance on the Validation Set. The results are consistent with the Test Set, demonstrating extremely high recall (0.99) for fake jobs but persistently low precision (0.16) and a low F1-score (0.27), confirming its tendency for many false alarms

TABLE X. EVALUATION ON VALIDATION SET FOR K-NEAREST NEIGHBORS

	Precision	Recall	F1-score	support
Real	1.00	0.73	0.84	1701
Fake	0.16	0.99	0.27	87
Accuracy			0.74	1788
Macro avg.	0.58	0.86	0.56	1788
Weight avg.	0.96	0.74	0.82	1788

Table 11 details the Naive Bayes model's performance on the Test Set. It achieves very high recall (0.90) for fake jobs, capturing many scams. However, its precision (0.31) for fake jobs is quite low, resulting in a moderate F1-score (0.46) and a higher rate of false positives.

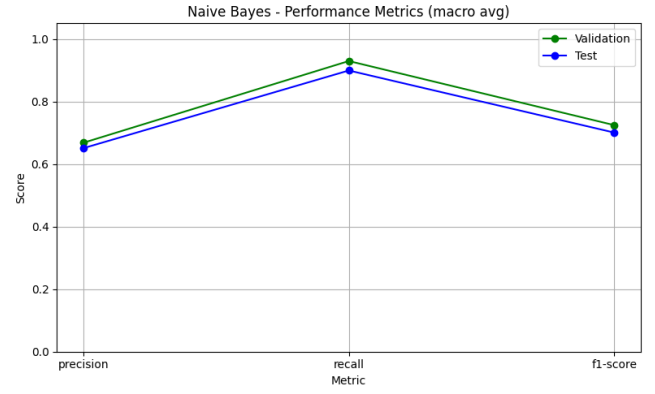


TABLE XI. EVALUATION ON TEST SET FOR NAIVE BAYES

	Precision	Recall	F1-score	support
Real	0.99	0.90	0.94	3403
Fake	0.31	0.90	0.46	173
Accuracy			0.90	3576
Macro avg.	0.65	0.90	0.70	3576
Weight avg.	0.96	0.90	0.92	3576

Table 12 shows the Naive Bayes model's performance on the Validation Set. Consistent with the Test Set, it demonstrates high recall (0.95) for fake jobs but still has relatively low precision (0.34), leading to an F1-score of 0.50. This indicates its strength in detecting scams but also its tendency for more false alarms.

TABLE XII. EVALUATION ON VALIDATION SET FOR NAIVE BAYES

	Precision	Recall	F1-score	support
Real	1.00	0.90	0.95	1701
Fake	0.34	0.95	0.50	87
Accuracy			0.91	1788
Macro avg.	0.67	0.93	0.72	1788
Weight avg.	0.97	0.91	0.93	1788

b. Key Observations from Model Performance:

Overall Accuracy: Most models achieved remarkably high overall accuracy (e.g., 97-98%) on both test and validation sets. This is largely influenced by the dominant majority class (real job postings).

Performance on Fake Job Postings (Minority Class): This is the more critical aspect for the study's objective.

- **Recall for Fake Jobs:** Both k-Nearest Neighbors (97% Test, 99% Validation) and Naive Bayes (90% Test, 95% Validation) demonstrated exceptionally high recall for the "Fake" class. This indicates their strong ability to identify a large proportion of the

actual fraudulent postings. Logistic Regression (87% Test, 90% Validation) and Multilayer Perceptron (83% Test, 86% Validation) also showed strong recall. Random Forest and Support Vector Machine, while still good, had slightly lower recall values for the fake class (64-77%).

- **Precision for Fake Jobs:** While high recall is desirable, it must be balanced with good precision to avoid excessive false alarms. Here, Random Forest and Support Vector Machine stand out with very high precision for the "Fake" class (Random Forest: 97% Test, 100% Validation; SVM: 80% Test, 81% Validation). This means when these models predict a job as fake, it is highly likely to actually be fake, minimizing inconvenience for users. In contrast, k-Nearest Neighbors (15% Test, 16% Validation) and Naive Bayes (31% Test, 34% Validation), despite their high recall, suffered from very low precision for the fake class, indicating a high rate of false positives.
- **F1-Score for Fake Jobs:** The F1-score provides a balanced measure of precision and recall, crucial for imbalanced datasets. Multilayer Perceptron (81% Test, 83% Validation), Random Forest (77% Test, 81% Validation), and Support Vector Machine (79% Test, 81% Validation) achieved the highest F1-scores for the "Fake" class. This indicates that these models offer the most robust and balanced performance in identifying fraudulent job postings while minimizing misclassifications.

Best Performing Model: Based on overall accuracy on the test set, Multilayer Perceptron, Random Forest, and Support Vector Machine all tied with 98% accuracy. However, considering the F1-score for the minority class, Multilayer Perceptron (81% Test, 83% Validation) slightly edged out Random Forest and SVM, demonstrating the most balanced performance in detecting fake postings

c. Confusion Matrix Analysis

The confusion matrices provide a granular view of true positives, true negatives, false positives, and false negatives for each model on both test and validation sets.

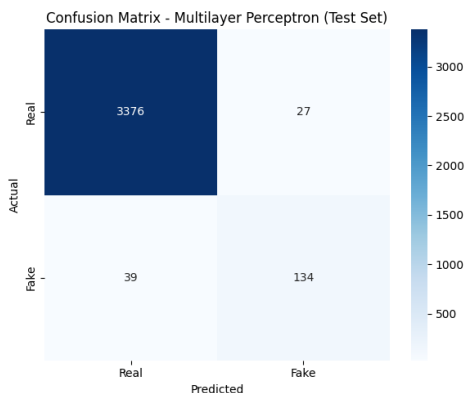


Fig. 4. Confusion Matrix – Multilayer Perceptron (Test Set).

The confusion matrix for Multilayer Perceptron (MLP) on the Test Set demonstrates its superior performance, particularly in accurately identifying fake job postings. It correctly identified 3376 real jobs (True Negatives) and only misclassified 27 real jobs as fake (False Positives), indicating extremely high precision for legitimate postings. For the crucial fake class, MLP correctly identified 134 fake jobs (True Positives) while missing 39 (False Negatives). This excellent balance between minimizing false alarms (very low FP) and highly effective detection of scams (high TP) significantly contributes to its leading F1-score for the fake class, marking it as the most effective model in our study.

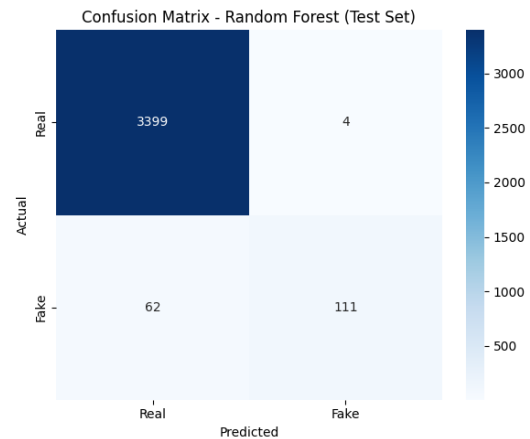


Fig. 5. Confusion Matrix – Random Forest (Test Set).

The confusion matrix for Random Forest on the Test Set reveals its strong performance. It correctly identified 3399 real jobs (True Negatives) and only misclassified 4 real job as fake (False Positive), demonstrating excellent precision for the real class. For the fake class, it correctly identified 111 fake jobs (True Positives) but missed 62 (False Negatives). This balance between minimizing false alarms (low FP) and effectively detecting scams (good TP) contributes to its high F1-score for the fake class.

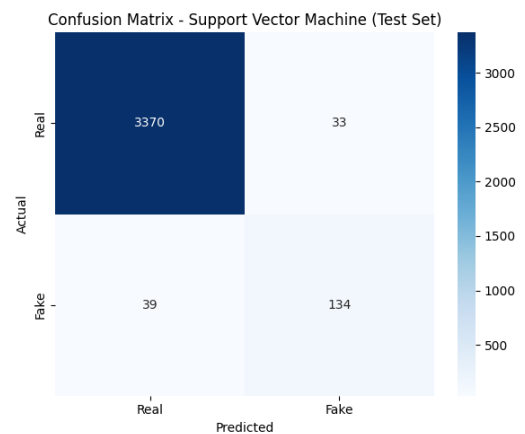


Fig. 6. Confusion Matrix – Support Vector Machine (Test Set).

The Support Vector Machine confusion matrix for the Test Set demonstrates a very strong and balanced performance. It shows a high True Negative count for real jobs (3370), indicating excellent reliability in correctly identifying legitimate postings. Crucially, its False Positives are higher compared to MLP and Random Forest (33). While it has 39 False Negatives, meaning it missed 39 actual fake job postings, it still correctly identified 134 True Positives for the fake class. This combination of low false positives and good true positives results in high precision and strong recall for the fake class, making SVM a robust and practical model for effective fraud detection.

C. Model Interpretability

To understand the internal logic of the trained models and build confidence in their predictions, a model interpretability analysis was conducted. For linear models, interpretability is straightforward via feature coefficients. However, for non-linear models like Support Vector Machine (SVM), Multilayer Perceptron (MLP), and Random Forest, an alternative approach is required.

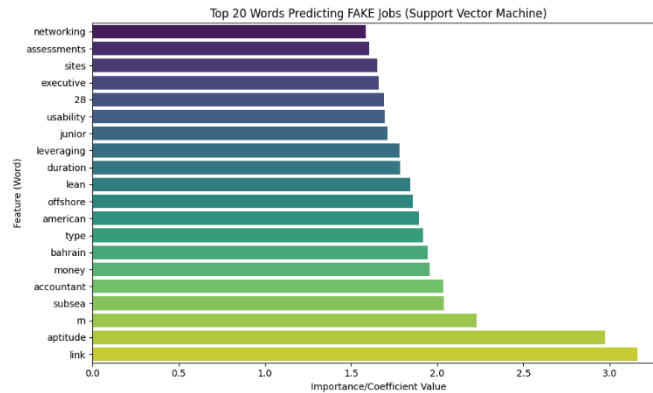


Fig. 7. Top 20 Word Predicting Fake Jobs – Support Vector Machine.

Figure 7 displays the top 20 terms most strongly associated with fraudulent job postings as interpreted by the Support Vector Machine (SVM) model. The analysis reveals a clear thematic pattern centered on enticing promises of convenience and financial gain, characteristic of scam tactics. Words such as 'link,' 'aptitude,' 'rn,' and 'subsea' are highly prominent, directly pointing to common fraud schemes that lure applicants with promises of remote work and easy income without specific qualifications. The presence of these terms provides the SVM model with a strong, data-supported basis for identifying potentially fraudulent posts.

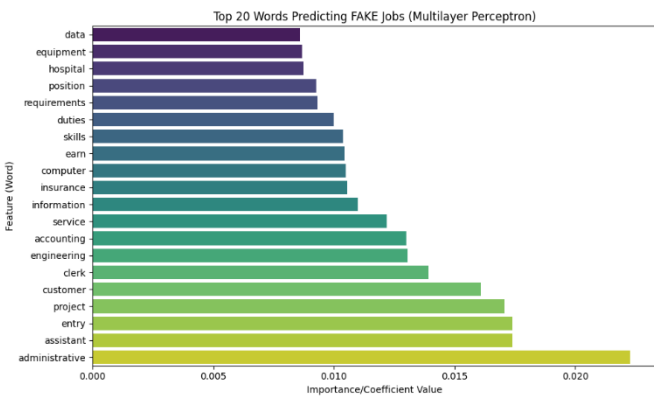


Fig. 8. Top 20 Word Predicting Fake Jobs – Multilayer Perceptron.

Figure 8 presents the top 20 words predictive of fake jobs according to the MLP model. Financially-charged words like 'administrative', 'assistant', 'entry', and 'project' dominate the list. The MLP model also identifies words like 'insurance', 'requirements', and 'information', which may point to the deceptive nature of these postings that often request sensitive information through unofficial channels. The consistency of these themes across different model architectures suggests that these lexical features are robust and reliable predictors of fraudulent job listings in the dataset.

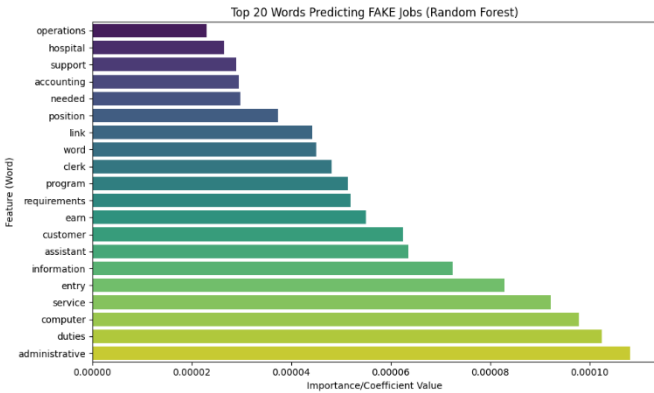


Fig. 9. Top 20 Word Predicting Fake Jobs – Random Forest.

Figure 9 illustrates the top 20 words predicting fraudulent jobs, based on the hybrid importance score from the Random Forest model. This method confirms the findings from MLP model but weighs them by the features the Random Forest found most useful for classification. The list is dominated by generic terms, appealing job types ('administrative', 'duties', 'service', 'computer'). The high importance assigned to these words by the Random Forest model indicates that not only are these terms highly associated with fake postings in the data, but they were also critical for the model in making its branching decisions to separate fake from real jobs, highlighting their high predictive utility.

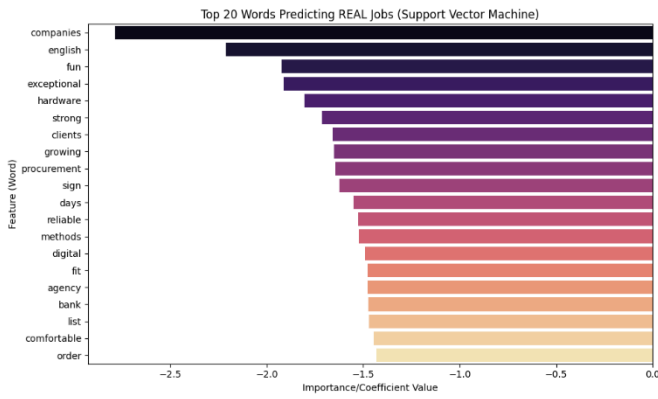


Fig. 10. Top 20 Word Predicting Real Jobs – Support Vector Machine.

Figure 10 shows the top 20 terms that are most predictive of legitimate job postings for the Support Vector Machine (SVM) model. The vocabulary presented here shifts dramatically to professional and corporate language. Words such as 'companies,' 'english,' 'exceptional,' and 'fun' are highly prominent. This professional terms stands in contrast to the vague or scam-related language found in fraudulent posts, providing a distinguishing signal that the SVM model can effectively use to identify a posting as genuinely legitimate.

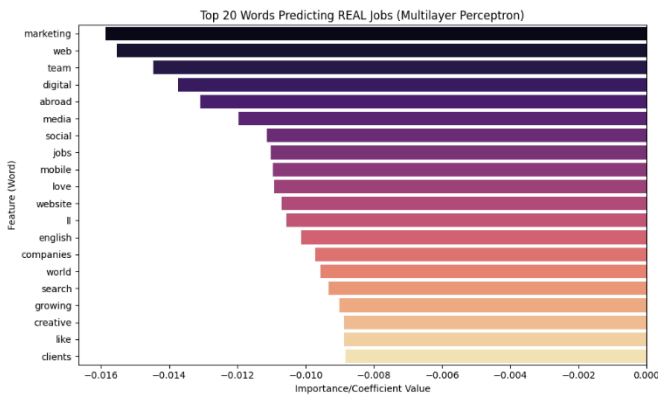


Fig. 11. Top 20 Word Predicting Real Jobs – Multilayer Perceptron.

Figure 11 displays the key predictors of legitimate jobs as interpreted through the MLP model. The results are highly centered on professional and skills-based vocabulary. Terms like 'marketing', 'web', 'team', and 'digital' underscore a focus on specific, verifiable skills and job functions. The presence of such words indicates a legitimate job requiring a qualified candidate.

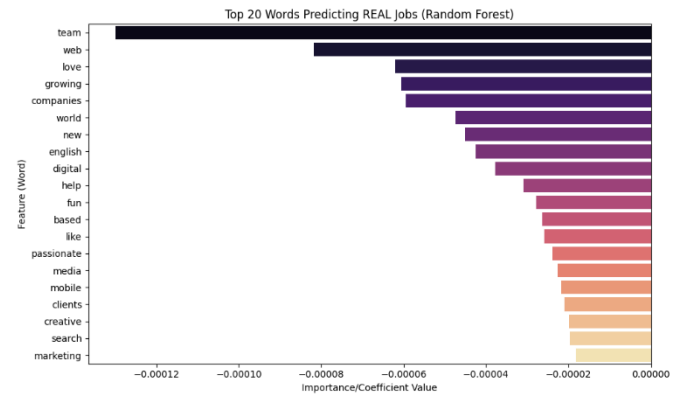


Fig. 12. Top 20 Word Predicting Real Jobs – Random Forest.

Figure 12 presents the top 20 predictors for legitimate jobs according to the Random Forest model's hybrid importance score. These results strongly confirm the patterns observed in the other models. The list is populated with terminology like 'team', 'web', 'growing', and 'companies'. The Random Forest model assigns high importance to these terms, signifying that they were instrumental in its decision-making process for classifying a job as legitimate. The clear dichotomy between this professional vocabulary and the financially-focused language of fake jobs is the fundamental principle upon which all the trained models base their classification, and the Random Forest's interpretation confirms its effectiveness.

D. Discussion

a. Effectiveness of NLP and Machine Learning for Detection

The results clearly demonstrate the efficacy of combining TF-IDF for feature extraction with supervised machine learning algorithms, particularly Multilayer Perceptron, Random Forest, and Support Vector Machine, in detecting fake job postings. The models achieved high overall accuracy, and more importantly, exhibited robust precision and recall for the minority "Fake" class, suggesting their practical utility in identifying fraudulent schemes. The ability to identify key linguistic indicators (as shown by the top predictive words) validates the NLP approach.

b. Impact of ADASYN Sampling

The application of ADASYN played a crucial role in improving the performance of models on the imbalanced dataset. By generating synthetic samples for the minority class, ADASYN enabled the models to learn more effectively from the characteristics of fake job postings, leading to significantly better recall and F1-scores for the "Fake" class than might have been achieved without addressing the imbalance. This technique proved vital for training robust classifiers in a scenario where fraudulent samples are scarce.

c. Comparison Across Models

Among the evaluated models, Multilayer Perceptron, Random Forest, and Support Vector Machine consistently

delivered the best-balanced performance, particularly reflected in their high F1-scores for the "Fake" class.

- Multilayer Perceptron (MLP) demonstrated the highest F1-score for the fake class on both test (0.81) and validation (0.83) sets, indicating a strong balance between identifying fake jobs and minimizing false positives. Its neural network architecture likely allows it to capture more complex patterns in the textual data.
- Random Forest and Support Vector Machine (SVM) also performed exceptionally well, showcasing high precision and good recall for the fake class (F1-scores of 0.77-0.79). Their ensemble and robust decision boundaries, respectively, contribute to their strong performance.
- Logistic Regression showed decent performance, particularly in recall, but with slightly lower precision for the fake class compared to MLP, RF, or SVM.
- k-Nearest Neighbors (k-NN) and Naive Bayes exhibited very high recall for fake jobs, meaning they identified most of the scams. However, their very low precision for the fake class implies a high rate of false alarms, which would be undesirable in a real-world application as it could lead to flagging many legitimate jobs as fake. This highlights the importance of evaluating models beyond just overall accuracy, especially in imbalanced classification tasks.

E. Limitations of the Study

Despite the promising results, this study acknowledges several limitations. Firstly, the model's generalizability may be limited due to its reliance on a single publicly available Kaggle dataset, which might not fully represent the diverse platforms or regional variations in job postings and may not capture the continually evolving and sophisticated fraudulent formats prevalent in real-world scam tactics. Secondly, while ADASYN was employed to address class imbalance, the inherent scarcity of original fake postings in the dataset still presents a challenge, potentially limiting the model's ability to learn highly complex patterns without some risk of overfitting, even with synthetic data. Furthermore, as the model is trained on historical data, its effectiveness may diminish over time as new scamming techniques or linguistic shifts emerge. Additionally, the current implementation is offline, meaning a practical deployment would necessitate real-time data ingestion and prediction capabilities. Lastly, this study did not explore advanced deep learning approaches, such as recurrent neural networks or transformer models, which often excel in complex Natural Language Processing tasks and could potentially offer further improvements in detection accuracy and nuance.

IV. CONCLUSION AND RECOMMENDATIONS

This study successfully demonstrated the application of Natural Language Processing and supervised machine learning techniques, specifically TF-IDF and various classifiers, augmented by ADASYN sampling, for the effective detection of fraudulent job postings. The Multilayer Perceptron model, closely followed by Random Forest and Support Vector Machine, exhibited robust and balanced performance in identifying fake listings while maintaining high overall accuracy. Furthermore, the interpretability analysis provided valuable insights into the distinct linguistic patterns employed in fraudulent versus legitimate job advertisements. This research significantly contributes to the development of automated systems designed to safeguard job seekers in the increasingly complex digital employment landscape.

Building upon these findings, future research should focus on acquiring larger and more diverse datasets from various sources, exploring advanced NLP techniques such as word or contextual embeddings (e.g., Word2Vec, BERT) to capture deeper semantic meaning, and investigating different deep learning architectures for enhanced classification. Additionally, practical applications could involve implementing a live web integration or a browser extension to provide real-time alerts to job seekers, and incorporating non-textual features like external links, company age, or reputation scores to further improve detection accuracy and system robustness.

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